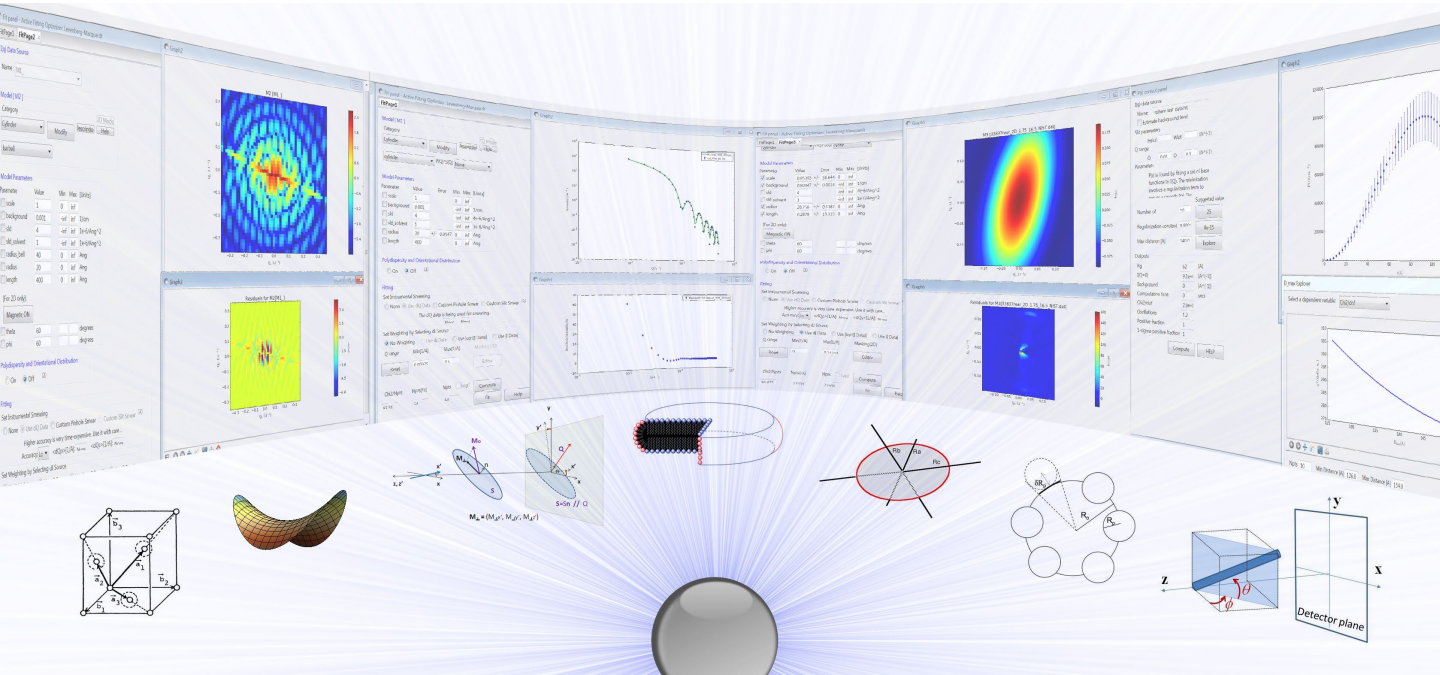
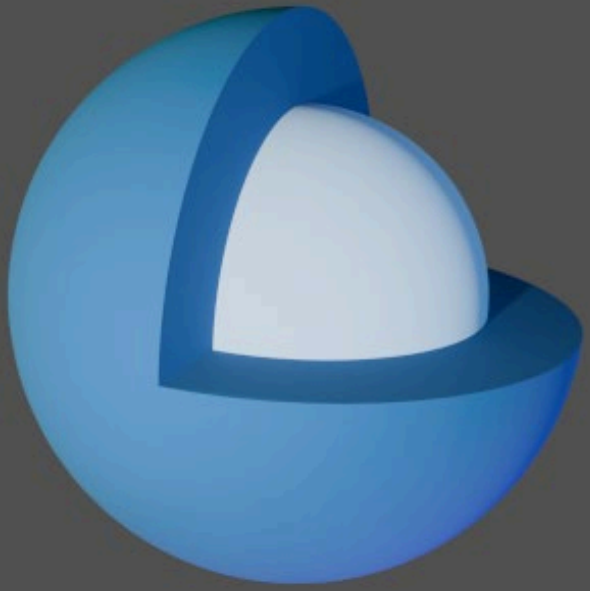
SasView and community software development in the age of Agentic AI



What is SasView and What Can it Do Currently?



SASVIEW

SMALL ANGLE SCATTERING ANALYSIS TOOLKIT

Data Analysis in SasView
What does that mean?

Only works on Reduced data

(All the instrumental artifacts are removed and only the science is left)

…. Sorta

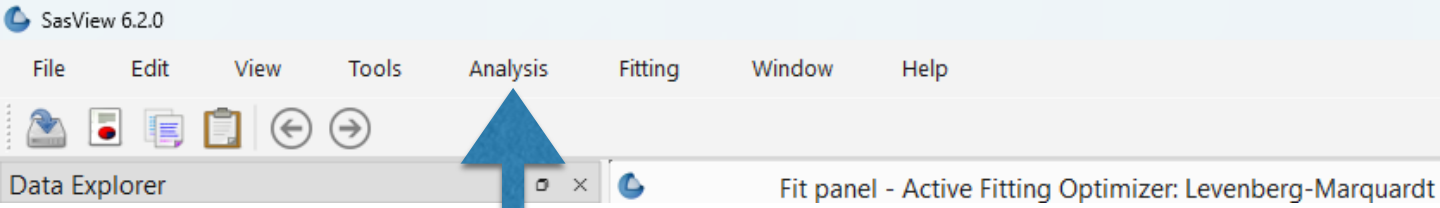
Focus on analytical approaches for this package

…. Sorta

Whatever anybody puts into it

…. Sorta

Perspectives on the data



Tools

- Data Operation
- File Converter
- SLD calculator
- Density/Volume calculator
- Slit Size Calculator
- Kiessig Thickness Calculator
- Q Resolution Estimator
- MuMag Fitter
- Generic Scattering calculator
- Shape2SAS Calculator
- Image Viewer
- Orientation Viewer
- Python Shell/Editor

Analysis

- Correlation Function
- Fitting
- Invariant
- Pr Inversion
- Size Distribution

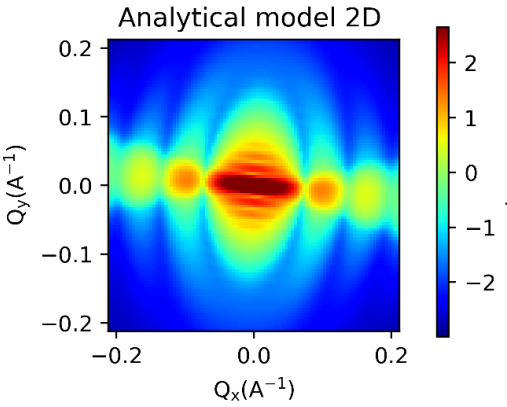
Plotting

DATA

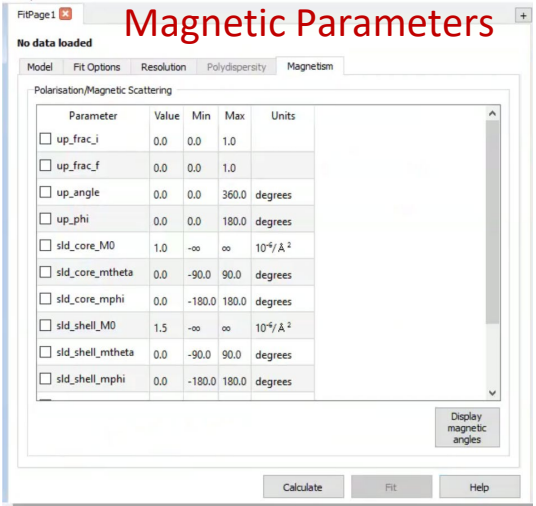
Current Capabilities

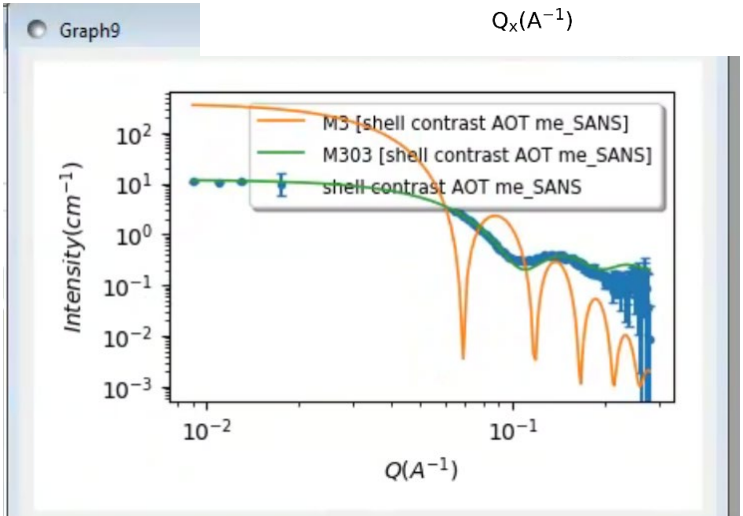
Data Management

- Most 1D formats and NXcanSAS 2D format suppor
- Some image and odd data supported through conv
- Data output in NXcanSAS or csv formats.
- Calculated curves associated with data (P(Q), S(Q), desmeared model, etc) accessible individually.
- Project saving capabilities (currently a bit fragile)

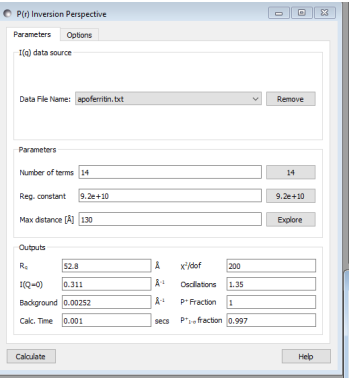


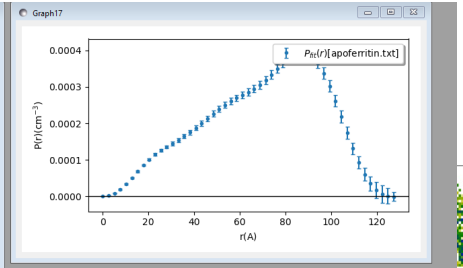
1D and 2D analysis

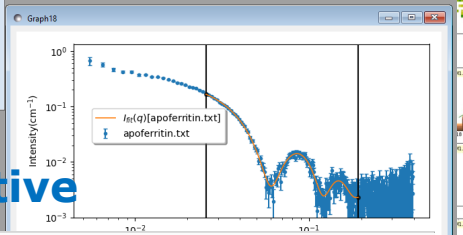




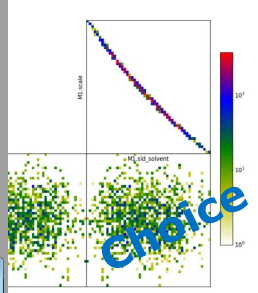
Current Capabilities

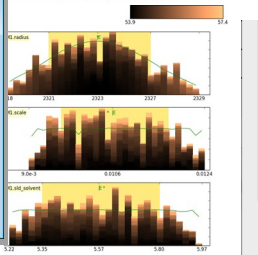




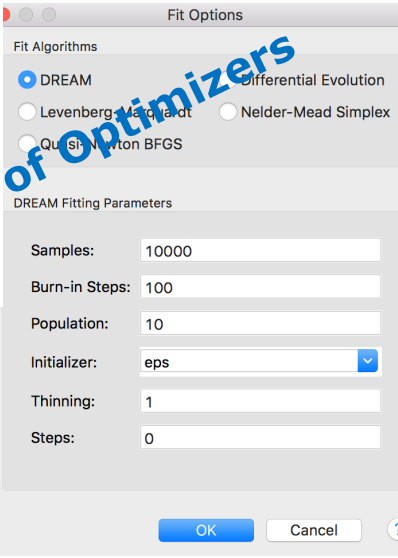


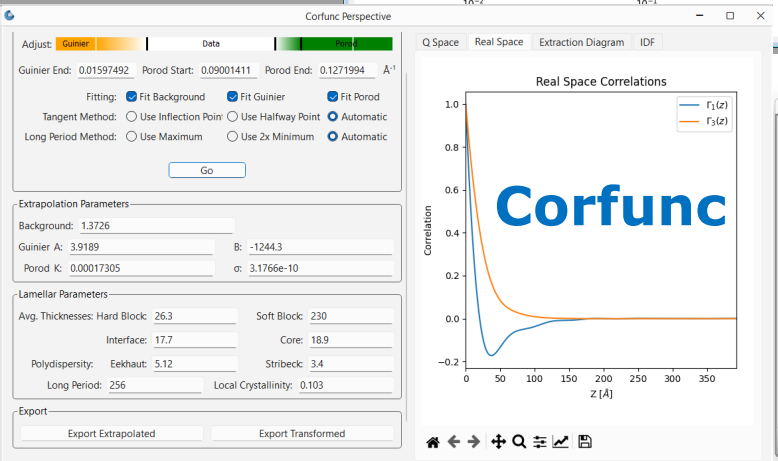
P(R) Perspective



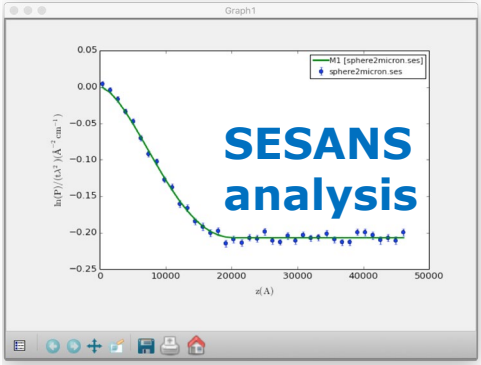


choice of Optimizers





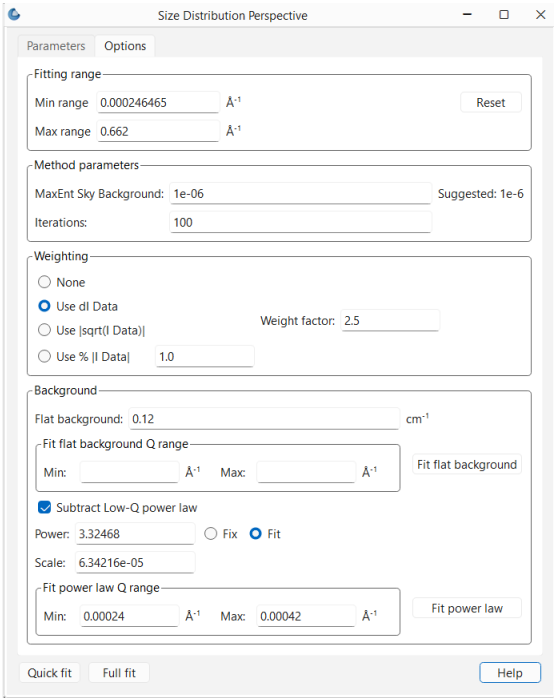
Corfunc

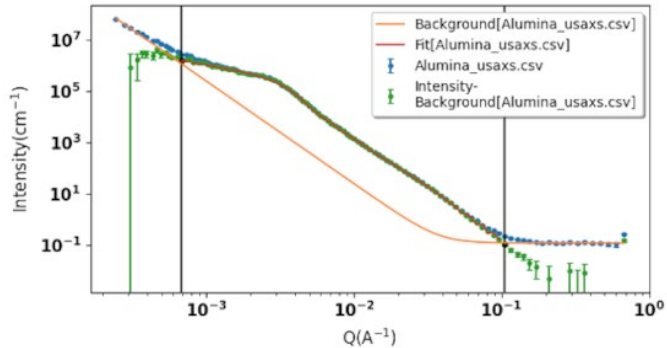


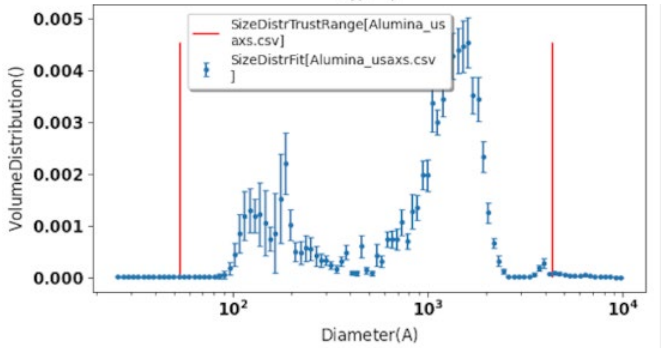
SESANS analysis

Current Capabilities: Size Distribution Analysis

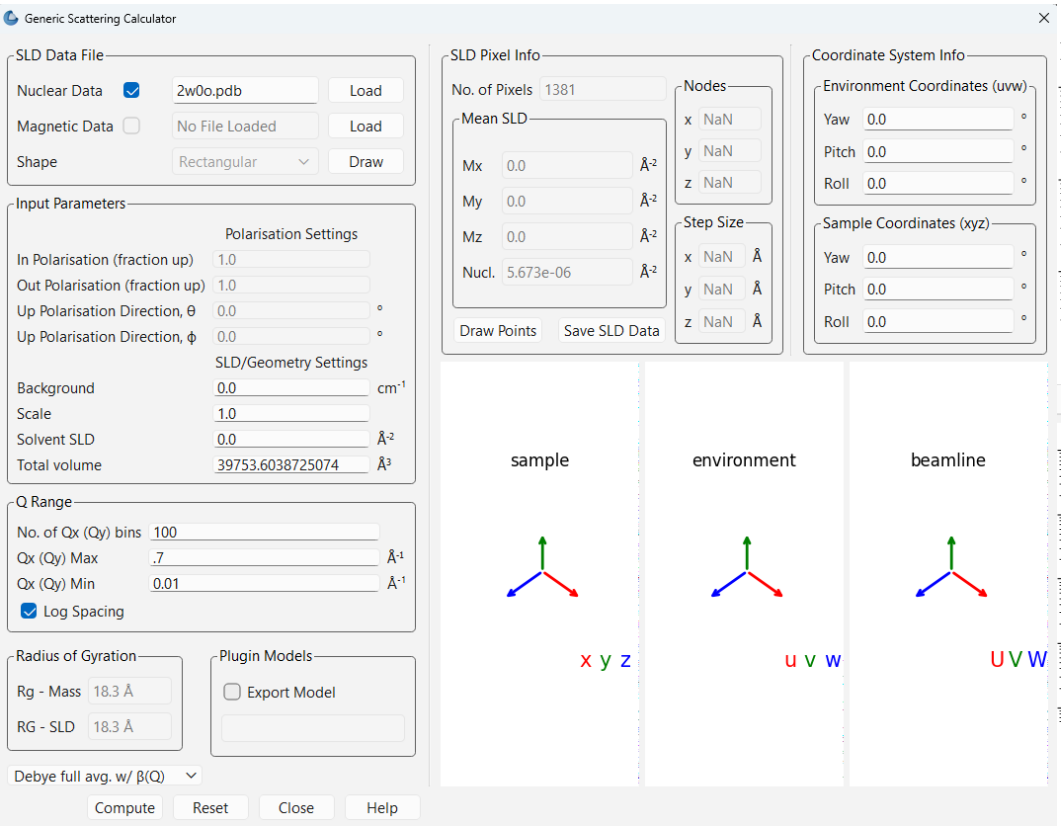
Based on Irena and using GSAS implementation of MaxEnt

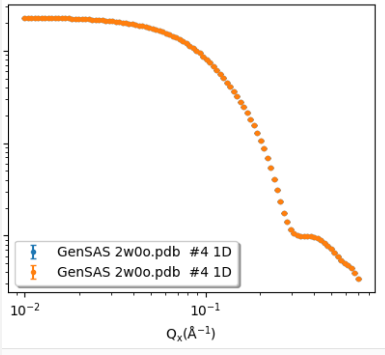


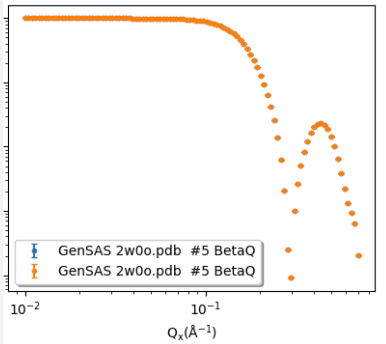




Current Capabilities: Generic Scattering Calculator







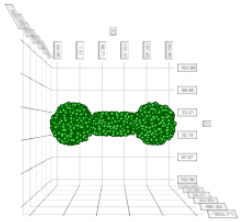
Current Capabilities: Shape2SAS

Shape2SAS (Experimental)

Build Model | Virtual SAXS Experiment

Add | hollow_sphere | Delete column | 3 | Exclude overlap

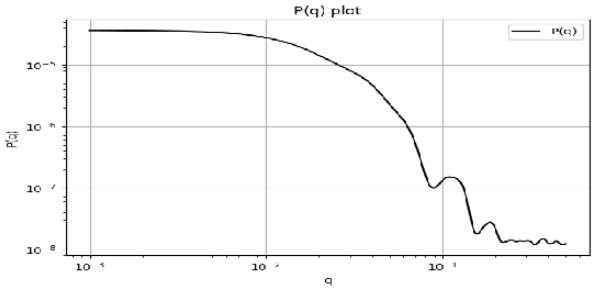
	1	2	3
1	sphere	cylinder	hollow_sphere
2	ΔSLD1 = 1.0	ΔSLD2 = 1.0	ΔSLD3 = 1.0
3	R1 = 50.0Å	R2 = 50.0Å	R3 = 50.0Å
4		l2 = 50.0Å	r3 = 25.0Å
5			
6	COMX1 = 0.0Å	COMX2 = 100.0Å	COMX3 = ...
7	COMY1 = 0.0Å	COMY2 = 0.0Å	COMY3 = 0.0Å
8	COMZ1 = 0.0Å	COMZ2 = 0.0Å	COMZ3 = 0.0Å
9	RPX1 = 0.0Å	RPX2 = 0.0Å	RPX3 = 0.0Å
10	RPY1 = 0.0Å	RPY2 = 0.0Å	RPY3 = 0.0Å
11	RPZ1 = 0.0Å	RPZ2 = 0.0Å	RPZ3 = 0.0Å
12	α1 = 0.0°	α2 = 0.0°	α3 = 0.0°
13	β1 = 0.0°	β2 = 0.0°	β3 = 0.0°
14	γ1 = 0.0°	γ2 = 0.0°	γ3 = 0.0°
15	Green	Green	Green



XY | YZ | ZY

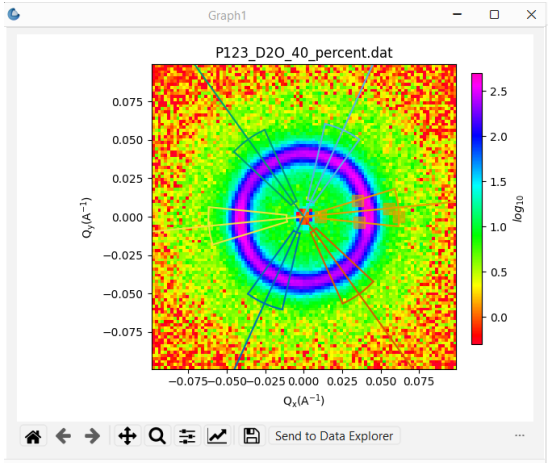
View radi 99.9 Å

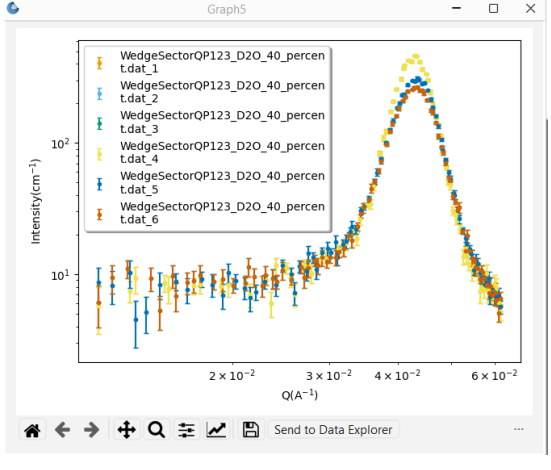
Scattering of P(q)

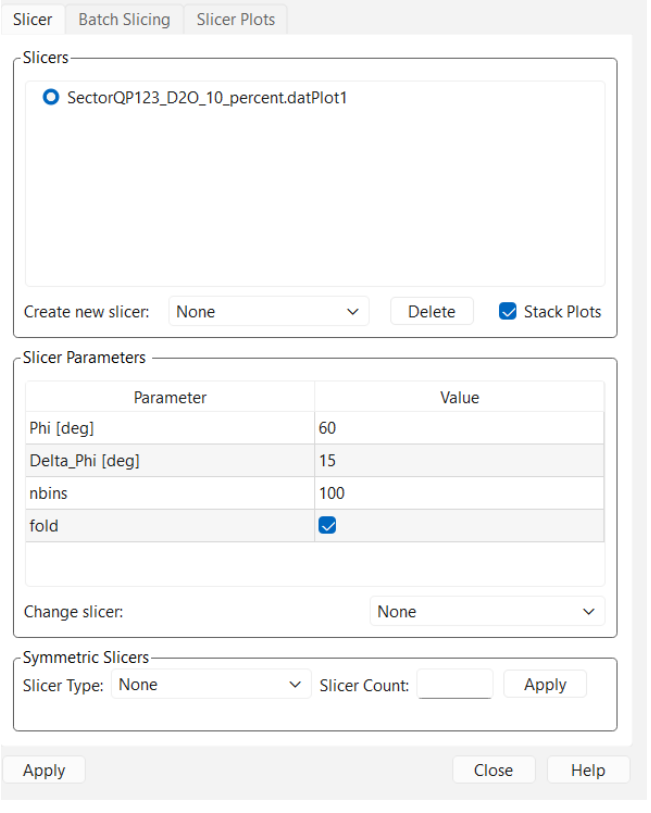


Plot | Include Scattering | Reset | Close | Help

Current Capabilities: Slicer Upgrade





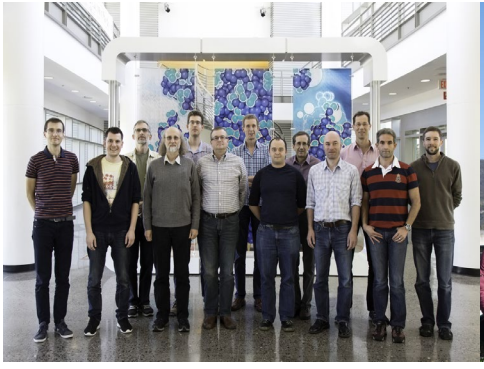


Current Activites/Refactoring

- New SVD approach to tracking reaction kinetics (MAXIV and SciLife Lab)
- Generic Scattering Calculator Refactor as Workflow (SciLife Lab and Diamond leads)
- SA(N)D Workflows - GRASP and GRIP functionality; Q_z; etc. (NCNR, J. Hopkins)
- **Data Structure and Plotting (ISIS leading and heavy lift)**
- Models (Jan Skov, Marianne Imperor, NCNR)
- Automation of routine tasks like releases (aspiration)

History

- 2006; originates in NSF *DANSE* project
- **2013; transitions into a community project**
- 2015; move to github and rename to SasView
- 2016; Sine2020 project funded by EU
- 2020 to present; Essentially a "volunteer army" with students and maybe most of an FTE

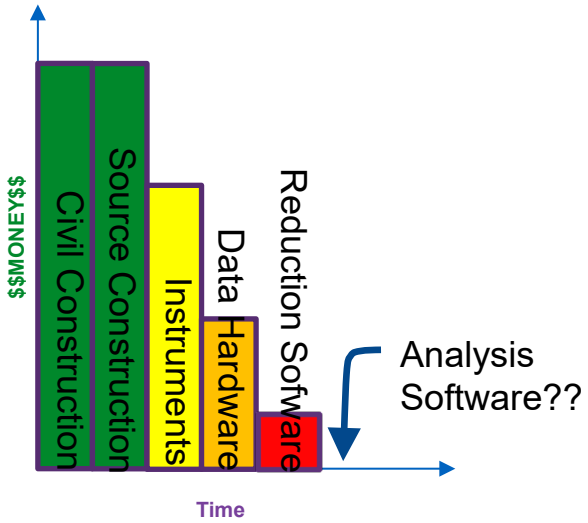




The Resource Problem

Analysis Software - Who's Job is it Anyway?

- The Facility?
- The User?
- The domain Science funding agency?



The unbounded problem

Fundamentally all these elements are relatively well defined problems…

EXCEPT …

Analysis Software is really unbounded. The needs are nearly infinite and constantly evolving.

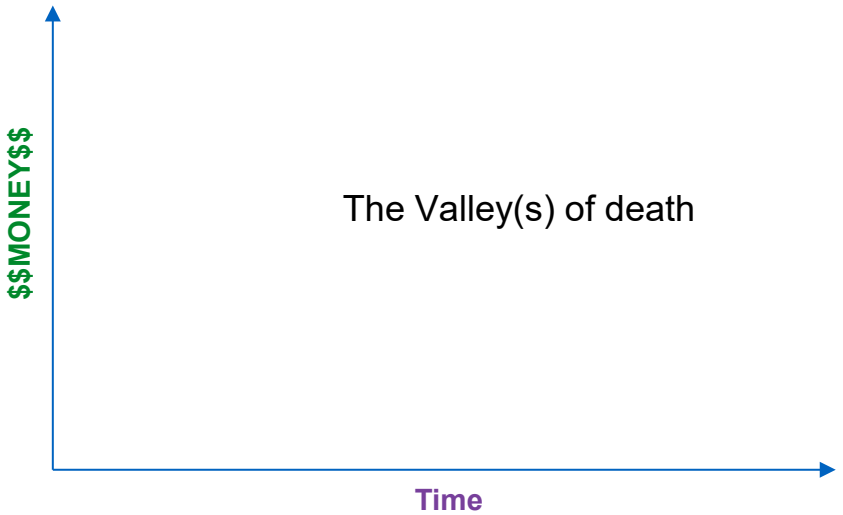
HOWEVER --- Analysis as defined here is also uniquely universal and ripe for collaborative pooling of finite resources

…. But beware the monoculture?

The Resource Problem

The feast and famine roller coaster

Analysis Software - Who's Job is it Anyway?



Special funding (grants) do not fund long term maintenance and ongoing development. They fund "big new (transformative) ideas"

The SasView Solution

vertical and horizontal **COLLABORATIONS**

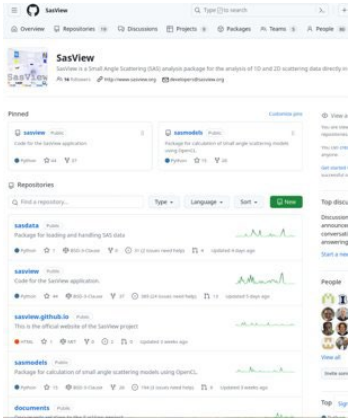
- **Facilities** provide foundational support through participation of data and instrument scientists
- **Grants and other projects** provide "bold new functionality"
- **Community**, writ large, helps provide support and functionality
- New ideas tested and developed as before by individuals or larger groups (the community)
- Once validated and deemed ready for the larger community these groups provide resources (their labor) to integrate while active developer community helps with training on where things go and on parts of interest to them (collaborative)

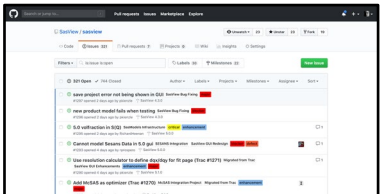
The
Open, Collaborative, Community Development
Model

Open, Collaborative, Community **Development**

Code is open source and publicly hosted at Github
Released under BSD 3-clause license

Code Hosting, Issue Tracking, Developer Wiki & CI on Public Github repos

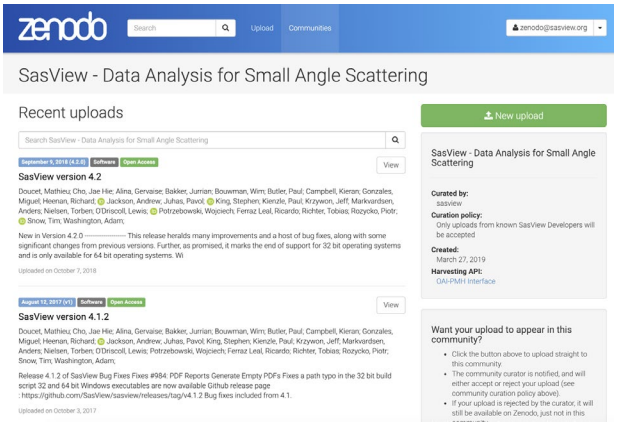




https://github.com/SasView

Rolling **5 Year Roadmap**

(Zenodo) DOI for **each release**



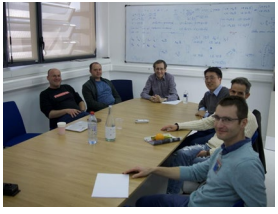
Open, Collaborative, Community Development





- Twice monthly zoom calls
- Regular ‘camps’ & ‘hackathons’
- Developer's mailing list
- SasView slack
- Expertise sharing and helping
- Small leadership team to facilitate

Collaboration also builds Community





http://www.sasview.org

http://github.com/SasView

Open, Collaborative, Community Development

Ask not what the community is going to do for you,
ask what you can do for the community

- P. Butler, March 2019

No MOU ... all are invited and welcome

Two Basic "Rules"

He who pays the piper gets to choose the tune…

Those who bring the resources (time and effort, or funds to buy time and effort) choose what to work on.

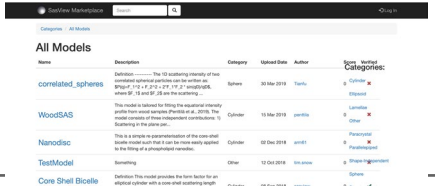
And ...

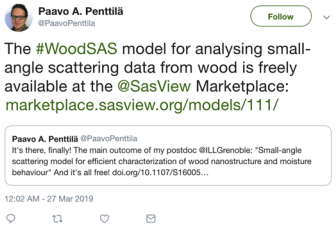
You cannot break existing experiences ...

- New dependencies vs long term maintenance (sustainability)
- Code quality vs long term sustainability
- Changing/degrading the current user experience for the existing user base

http://marketplace.sasview.org

Model Marketplace for Users to share their models





What will SasView do in the future?



Whatever the community contributes.

Current Status

- >100 contributors from >30 organizations, including 20 Universities!
  (~10-15 active at any one time- at best)
- 1 to 2 releases/year
- Documentation/tutorial projects ongoing
- Usage? Seems to be "everywhere?"
- Lots of "compliments" (a bit scary)
- 210 Publications in 2025



SasView Model: Some Current Challenges

- The tragedy of the commons: somebody else will take care of the problem – New features are more interesting than fixing fragility issues
- Hard to get non-coders to believe they can contribute equally – it is unfortunately in the name "code camp" … words matter → "Contributor camp"
- Building community, especially during a pandemic is HARD WORK
- Lack of funds for small things … but beware too much money?

- Cost of writing sustainable and maintainable code is surprisingly high and can be a deterrent to new contributors
- Hard to keep up with increasing security issues with the resources so far.

The price of success:

- Many people view the project as a well-funded group of professionals → barrier to contribution
- Large project with many moving pieces is a barrier to new volunteer coders

AI to the RESCUE!

- Lower the barrier to entry considerably!
- Democratization? – bench scientist can take more control?
- Helps understanding code structure (Munich camp talk)
- Help solve issues
- Write new features?!!



AI to the RESCUE!

Solved the Riemann Hypotheses! What can it NOT do?

- Swarms of agents synthesized results from various papers

Super intelligence is on the horizon!! WOOHOO!??? Or not?

NIST Director Newsletter:

> *I left the summit reflecting on a what-if question. What if we are all really underestimating the amount of uplift AI can have on our work? Many, including myself, interact with AI models as if they are very knowledgeable consultants. But the real unlock could come from agentic AI, when we put a harness around the AI models to persistently work on what we need them to do on a regular basis. In other words, agentic AI offers the possibility of each one of us having our own new, highly knowledgeable employees that work through our computers, that need to be trained and occasionally supervised.*

AI to the RESCUE!
What *is* Agentic AI?

My journey

Updating the SAS portal

www.smallangles.org

https://dev.smallangles.org

On a burner server

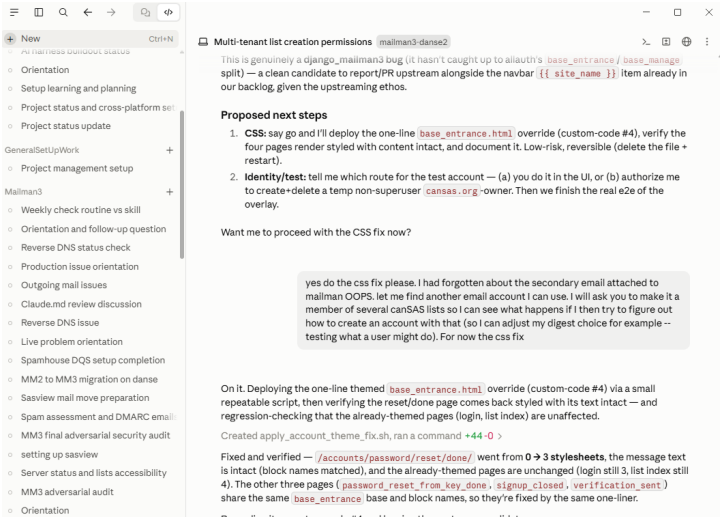
Building new mailing lists

https://lists.sasview.org

https://lists.cansas.org

https://lists.smallangle.org

On production server



Fixing an issue from github

https://github.com/SasView/sasview.github.io/issues/33

AI to the RESCUE!
What *is* Agentic AI?

Not a magic wand

- Effort does not go away – it changes (and time needed ↘)
- There is a leaning curve – the sweet spot on utility
    - No technical knowledge/skill
    - Strong technical understanding/some skill
    - Technical experts/Very skilled
- Requires a careful planning phase
- Big projects will require iteration
- Managing context window is KEY (pea brain syndrome aka: the dumb zone)

Ignore this and you get SLOP

AI to the RESCUE?

PROBLEMS for open source software

- Moves the time balance from writing to reviewing code
- Too easy makes for lots of slop code. Lots of complaints from professional developers See above problem
- Open source projects report a huge uptick of contributions, mostly “slop” making it impossible to review and find the useful ones.

……. Maybe not our problem in SasView…..

AI to the RESCUE?

How are open source projects responding

- NO AI generated code!! (how enforced?)
- First PR must be without AI (need to understand the code base)
- Trust: person submitting with AI is recommended by an existing contributor
- LINUX approach. AI as helper but every submission is considered to be from the human author who is held responsible for the code.
- Don’t worry about it? … probably rapid code deterioration.
- If you can’t beat them join them?
- Provide basic context file (*.md) with code structure and delivery rules? More? Enforced?

AI to the RESCUE?

What about the OTHER side of the equation – the USER

How will the tool be used in a world of agentic AI?

- Vibe coding new models
- Wrapping SasView inside an agentic AI framework with documentation accessed as a RAG
- Asking Gemini for help?

AI to the RESCUE?

What about the OTHER side of the equation – the USER

How should developers respond?

- MCP connectors so users can roll their own?
- Build in better hooks for agents: e.g. markdown files with code layout and structure for CLI to hand your agent?
- Built in agents like is available on IDE and github etc?

SasView leadership (Wojciech) proposing a user survey…

AI to the RESCUE!...?

The elephant in the room: COST!!



Democratization?

- Who pays?
- Agentic AI calls to run sasview can use lots of tokens
- Agentic AI for development can use lots of tokens
- Developers using it effectively vs not..
- ... AI Data centers?

SasView and community software development in the age of Agentic AI

AI is here to stay and is evolving exponentially

In order to stay relevant we need to figure out how to harness its power without being destroyed (tame the beast)





Expect Discussions at contributor camp in Prague and welcome input here today!

DISCUSS!



